How the Choice of Sign Inventory Moves the Statistics of an Undeciphered Script

Rongorongo under three published readings, with the Indus script as an outside check

Ilpo Väätäinen*

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Abstract

Statistical arguments about undeciphered scripts, whether a sign system "behaves like language", are computed on a transliteration, and a transliteration presupposes a catalogue of signs. For rongorongo the published catalogues disagree by an order of magnitude, from about fifty basic signs to about six hundred. Pozdniakov and Pozdniakov [2007] observed that the statistics "will differ markedly depending upon the inventory of glyphs chosen"; the point has not, to our knowledge, been quantified.

On the CEIPP corpus of twenty-five objects and 14 812 glyphs, using the normalised conditional entropy of Rao et al. [2010], a standard measure of how predictable the next sign is given the one just before it, we tried three published readings of the *same* inscriptions: three different, defensible decisions about what counts as one sign. Just relabelling the same glyphs moved the statistic by 0.318 of its $[0, 1]$ scale. The effect the statistic exists to detect, how far the real corpus sits from a shuffled version of itself with all sequential order destroyed, is at most 0.077, that being the largest of the three readings; the smallest is 0.046. Changing the catalogue therefore moves the statistic about four times further than the effect it is meant to measure: most of what changes when you switch readings is not the script but the coding of it. A sweep over sizes from 25 to 633 shows this is mostly an effect of catalogue *size*, and that the field's own catalogue of about 125 signs sits where a rising distance from the null meets collapsing coverage. Composition matters less for entropy than for the scribes' own substitutions, where the published catalogue absorbs 40 % of variants against 18 % for the best mechanical rule of its size.

Against real syllabaries at matched size and length, a fifty-sign rongorongo reproduces the syllabary's *number* while carrying 1.6 to 5.5 times less sequential structure, a gap that survives matching of run length, sample, genre and orthography. The identical code run on the Indus script returns the opposite verdict (close catalogues, a large distance from the null), so this is a property of one script's transliteration problem, not of undeciphered writing in general. The levels of these statistics belong to the catalogue and the sample; only differences travel.

1 Introduction

Every quantitative claim about an undeciphered script rests on a transliteration, and every transliteration rests on a prior decision about what counts as one sign. For rongorongo, the script of Easter Island, that decision has been made several times with very different results. Barthel [1958] catalogued about 600 glyphs; the CEIPP transliteration inherits that inventory and adds variant letters; Horley [2021] reduces it to about 125 basic glyphs by decomposing ligatures and merging allographs (different-looking glyphs treated as the same underlying sign, the way capital *A* and lowercase *a* are one letter); Pozdniakov and Pozdniakov [2007] argue for

*Independent researcher, Helsinki. `ipezygj2@gmail.com`. Code, results and this draft: `https://github.com/ipezygj/rongorongo-catalogue-audit`, archived at doi:10.5281/zenodo.21964265.

a core of 52. These are not competing readings of a few doubtful glyphs. They are different answers to the question of how many symbols the script has, and the answers span a factor of ten.

The statistics used to argue that a sign system is, or is not, linguistic are all computed on the transliterated sign stream. They are fingerprints of predictability and structure: conditional entropy [Rao et al., 2009, 2010], its critics [Sproat, 2010, 2014], *n*-gram and Zipfian profiles (whether short sequences and sign frequencies follow the patterns real languages do), and hapax ratios (how many signs occur only once). Pozdniakov and Pozdniakov [2007] state the consequence plainly: "it is impossible to make use of statistical data, for clearly the data will differ markedly depending upon the inventory of glyphs chosen." The remark is qualitative, and it appears in a paper whose response to the problem was to build a better catalogue rather than to measure how much the choice moves the numbers. Davletshin [2012] goes further and says there are "no efficient sign catalogues based on a thorough analysis of contexts", holds both published inventories unsatisfactory for textual work, and uses descriptive nicknames in their place; Davletshin [2017] points out that Barthel himself gave three counts for one corpus (799 catalogue positions, 322 signs occurring at least three times, about 120 irreducible constituents) that cannot be reconciled with one another. Korovina [2026] runs the same *n*-gram method on two transliterations of the same inscriptions and obtains different verdicts, but treats the gap as a caveat rather than as the object of study.

This note treats the gap as the object of study. The question is not whether rongorongo is writing. It is: *when a statistic is computed on an undeciphered corpus, how much of the number is the corpus and how much is the catalogue?* We answer it on rongorongo because the catalogue disagreement there is unusually large and the corpus is unusually well digitised, and we answer it in the units in which the debate is now conducted, the length-normalised conditional entropy of Rao et al. [2010], so that the movement can be read directly against the distances that separate languages from DNA, music and code on their plot.

Our contribution is a set of measurements, made on public data with public code, and a reading of them that we have tried to keep within what the measurements support:

1. The catalogue moves the statistic about four times further than the effect it is meant to detect (§4).
2. Most of that movement is catalogue *size*, not composition; there is no preferred size below the published $\sim$125, and no "knee" (no sudden bend in the curve) at the syllable count (§5).
3. Composition matters, but for a different question: the published catalogue encodes graphic knowledge that no mechanical rule of the same size recovers, visible against the scribes' own substitutions rather than in conditional entropy (§6).
4. A small catalogue makes rongorongo *look* syllabic on the raw axis; on the null-relative axis it carries less sequential structure than every real syllabary at matched size and length (at least 1.5 times less, and 2.4–3.6 times less against Polynesian prose at its own size), even under its best catalogue (§7).
5. Transliteration error, sample size and estimator bias are all real, and all an order of magnitude smaller than the catalogue effect (§8).

2 Data

Corpus. The CEIPP transliteration as maintained and published in XML by Philip Spaelti[1], one file per object, one `<glyph>` element per coded glyph with a Barthel/CEIPP code (e.g.

[1] https://kohaumotu.org/Rongorongo/xml/

022bfy) and a link marker. 25 objects. Line breaks, lacunae, illegible glyphs and end-of-text markers cut the sign chain: a bigram is counted only when the object attests the adjacency. Under the numeric reading (variant letters stripped, ligatures split) the corpus has 14 812 tokens over 633 types in 1 182 uninterrupted runs.

We verified against the fixed-column sample CEIPP publishes on its own site that the widely used `rongopy` tablet files are this transliteration reformatted (15/15 glyphs of the published sample match). Provenance therefore reduces to CEIPP; the "readings" below are published *granularities* of one transliteration, not independent editions. That is a weaker comparison than three independent editions would be, and it is the one available.

Readings. Three levels that correspond to published positions on what counts as one sign:

- **full**: CEIPP codes with variant letters, ligatures split. $L = 1\,897$ types.
- **base**: numeric Barthel code only. $L = 633$.
- **horley**: Barthel → Horley basic-glyph map as shipped with `rongopy` (638 keys → 126 distinct values, 372 of the keys decomposing a ligature into parts). One of the 126 is "?", the map's own marker for no basic glyph, which leaves 125 signs. Codes the map does not reach, or sends to "?", break the chain like a lacuna. Coverage 91.2 % of coded glyphs. $L = 125$, 15 415 tokens (decomposition increases the count).

Reference syllabaries. (§7.) Rapa Nui: the New Testament (gospels and Acts, 117 chapters, Wycliffe translation), syllabified mechanically as (C)V with C ∈ {p t k ' m n ŋ v r h}, V ∈ {a e i o u}, long vowels either kept ($L = 105$) or folded ($L = 55$); 322 752 syllables. Māori: Paipera Tapu New Testament (ebible.org), same procedure, C ∈ {h k m n ng p r t w wh}, $L = 98$ / 52; 460 155 syllables. Linear B: the `linearb.xyz` corpus, 5 889 inscriptions, syllabograms only, word dividers dropped (rongorongo has none), ideograms and numerals cutting the chain; 47 364 signs, $L = 95$. All three are measured on twenty random contiguous windows of exactly the rongorongo token count so that coverage and null distance are compared at matched N. Which token count is not one number, and the tables below do not all use the same one: Table 3 matches the numeric reading's 14 812, Table 4 the 15 115 of the Horley reading on the objects of at least 100 glyphs (Appendix A), and the run-length check of §7 the numeric reading's 1 182 runs. The three differ by about two per cent in N and rather more in segmentation; each comparison is internally matched, and none is matched to the others. The Māori text is public domain; the Linear B corpus is taken from its public repository; the Rapa Nui text is copyright Wycliffe Bible Translators and is used for measurement only, fetched by the reader from its publisher and never redistributed.

Two codes that are not signs. CEIPP code 000, written `000!` in 380 of its 390 occurrences, marks a glyph that is present on the object but which the edition declines to identify. Code 999 is the vertical divider of the Santiago Staff: 97 tokens, all on that one object, and identified as a divider rather than a sign in Sproat's string-matching pages. The Horley reading drops both, because the map returns nothing for them and unmapped codes break the chain like a lacuna. The numeric and full-variant readings keep them unless told otherwise, so 000 enters those readings as the sixth commonest "sign" in the corpus. Treating both as breaks in every reading (`pseudoglyph_audit.py`) moves the spread across readings of §4 from 0.318 to 0.327 and the ratio from 4.14 to 4.28. The figures in the sections below are computed with the two codes kept, and are therefore conservative; the corrected pair is given here and again in §9.

Changing the reading changes more than the catalogue. Two things move with it besides the number of classes. Splitting ligatures raises the token count, so the Horley reading is 15 415 tokens against the numeric reading's 14 812 on the same carvings; and because codes the map cannot reach break the chain like a lacuna, Horley is cut into 1 771 runs of mean 8.7 where the numeric reading has 1 182 of mean 12.5. "Hold the corpus fixed and change only the catalogue" is therefore not exactly what happens: tokenisation and segmentation travel with the reading. The size of that is measurable rather than arguable, and it is the audit of the previous paragraph: treating the two non-sign codes as breaks in every reading equalises the worst of the segmentation difference and moves the ratio from 4.14 to 4.28. The effect the paper reports is larger, not smaller, once the readings are cut the same way.

3 Method

Two quantities recur through the rest of the paper, and the argument is a comparison between them, so we name them before the formulas. The first is the *statistic*: the number a given transliteration yields, here the normalised conditional entropy $\hat{H}_c$ defined below. The second is the *distance from the null*, written Δ: how far the real corpus sits from a shuffled version of itself with all sequential order destroyed, which is the effect the statistic exists to detect. In estimation terms the first is an estimate and the second is the estimand; nothing in this paper needs a term of its own beyond those. The method, in one sentence, is to hold the corpus fixed, change only the catalogue, and ask how far each of the two moves.

Statistic. A *unigram* is one sign counted on its own; a *bigram* is a pair of signs counted as they occur next to each other. Conditional entropy asks how much uncertainty is left about the next sign once you already know the one before it: a script where certain signs reliably follow others (the way q is almost always followed by u in English) has *low* conditional entropy, and a script where any sign could follow any other has *high* conditional entropy, close to what a shuffled, structureless sequence gives. For a sign stream with unigram counts c_i and bigram counts c_{ij} over L observed types, the plug-in conditional entropy is $H_c = H(\text{bigram}) - H(\text{unigram})$ in bits. Following Rao et al. [2010], who take the logarithm to base L "to compare sequences over different alphabet sizes," we report $\hat{H}_c = H_c / \log_2 L$, which lies in $[0, 1]$ between their Min Ent and Max Ent references. Every distinction their plot draws lives on this axis. Raw bits would make catalogue choice look damaging for free, because changing L changes the ceiling; the normalisation is what divides that out, and the question is whether the movement survives it.

Null. A unigram-matched shuffle: tokens permuted across the corpus, run lengths preserved, so the unigram distribution and L are identical and all sequential structure is destroyed. 200 permutations per reading (100 in the sweep, 30–50 in the reference runs). The distance from the null is $\Delta = \hat{H}_c(\text{real}) - \overline{\hat{H}_c(\text{null})}$; z is that difference over the null standard deviation: in plain terms, how many standard deviations the real corpus sits from what the 200 shuffles produced by chance. A z of, say, -70 means a gap that large would essentially never happen if the corpus really were structureless; it says nothing by itself about how *big* the gap is, only how sure we can be that it is not noise. A negative Δ means the corpus is more predictable than its shuffle.

Coverage. The fraction of the $L \times L$ bigram table with at least one observation. It says whether H_c is estimable at all at that L and N.

Merge rules (sweep). Starting from the 633 numeric codes, a catalogue of size L is produced by one of: *numeric*, sort by Barthel number and cut into L contiguous classes of equal type

reading	L	tokens	coverage	$\hat{H}_c$	Δ	z
full (CEIPP + variants)	1 897	14 812	0.25 %	0.385	−0.046	−70
base (CEIPP numeric)	633	14 812	1.74 %	0.522	−0.055	−68
horley (published ∼125)	125	15 415	23.9 %	0.703	−0.077	−67
spread across readings				0.318		
largest Δ of any reading					0.077	
ratio				4.1×		

Table 1: Three published granularities of the same 25 inscriptions on the Rao–Sproat scale. Δ is the distance from the unigram-matched null (§3); the spread is the range of $\hat{H}_c$ across the three readings. Miller–Madow correction applied to real and null alike gives spread 0.309, Δ 0.083, ratio 3.7×, and does not reorder the readings.

count (Barthel's numbering follows shape, so this is the cheapest shape-based rule); *horley*, the published 125 merged further downward by the same numeric cut; *freq*, keep the $L - 1$ most frequent codes and pool the rest; *random*, L classes of equal type count drawn at random (20 draws), which is what a catalogue of size L with no information in its composition does. Everything above the random line at a given L is what composition adds; the line itself is what size does.

Code. All scripts are standard-library Python and are released at `https://github.com/ipezygj/rongorongo-catalogue-audit` (archived: doi:10.5281/zenodo.21964265), with a fetch script for every input (nothing third-party is redistributed). Random seeds are fixed; every number below is reproducible from the public XML.

4 The catalogue moves the statistic four times further than the effect

This section asks the paper's question in its simplest form. Hold the carvings fixed, change nothing but the decision about what counts as one sign, and ask two things: how far does the statistic move, and how large is the effect it is supposed to be detecting? If the first is much larger than the second, then a published value of this statistic is a fact about the transliteration before it is a fact about the script.

Three scholars looked at the same twenty-five inscriptions and made three different, reasonable calls about what counts as one sign; nothing about the carvings changed, only the coding of them. Table 1 is the headline. Structure is present under every reading: the corpus is 67 to 70 standard deviations more predictable than its shuffle. What does not survive the change of reading is any statement about *how much.* Moving from the variant-level reading to Horley's moves $\hat{H}_c$ by 0.318 (nearly a third of the whole axis on which Rao et al. [2010] separate languages from non-linguistic systems), while Δ, the effect that movement has to be judged against, is 0.077. That denominator is itself a choice, and we take the one least favourable to the argument: 0.077 is the largest Δ any of the three readings reaches, and dividing by the smallest, 0.046, would give 6.9 rather than 4.1. The effect is not one number either, which is the paper's own point turned on its own ratio.

The sharpest single observation is in the coverage column. Under the variant reading 0.25 % of the bigram table is observed and the script looks highly structured (0.385); under Horley's 23.9 % is observed and it looks much closer to Max Ent (0.703). *The more estimable the statistic, the less language-like the script appears.* High "structure" under fine-grained readings is, at least in part, sparse-table bias, not a property of the writing.

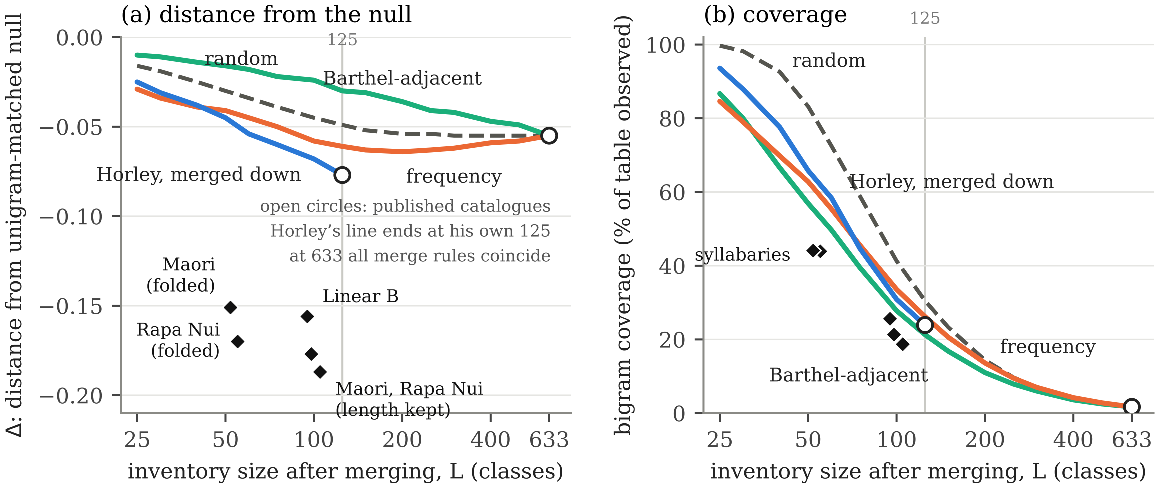

Figure 1: Rongorongo under four merge rules (lines) and real syllabaries at their own size (diamonds; Rapa Nui and Māori syllabified with vowel length folded or kept, and Linear B), all at the rongorongo token count. *The lines are merge rules, not catalogues.* One fixed stream of 14 812 glyph tokens is repartitioned into L classes by each rule and the statistic recomputed, so the horizontal axis is the granularity of a partition and not a choice among published sign lists: at $L = 633$, the CEIPP numeric codes, every rule is the identity and the four lines meet; below it they differ only in which codes are merged. The published catalogues are the open circles; Horley's line ends at his own 125, because his map cannot be un-merged past his inventory. Nothing is merged for the syllabaries, which is why each of them is a single point. (a) Distance of the corpus from its unigram-matched shuffle on the log-base-L scale: more negative is more sequential structure. Only the frequency rule has an interior optimum, at $L = 200$, and Horley's published 125 is the largest value his sequence reaches; every syllabary sits further from its shuffle than the best rongorongo catalogue at the same L, by 2.4–3.6 for the Polynesian texts shown. (b) Share of the $L \times L$ bigram table observed. The grey line marks $L = 125$.

Sample size. Rao et al.'s Indus corpus has about 7 000 sign occurrences over 417 types, 16.8 tokens per type; the CEIPP numeric reading has 23.4. Rao et al. compute block entropies over n-gram orders and note that "beyond $N = 6$ the entropy estimates become less reliable due to the small sample size"; their N is the block length, not the token count that N denotes everywhere else in this paper. On random sub-samples of the present corpus, H_c is still rising at 100 % of the data, and under the variant reading the last quarter of the corpus contributes 83 % of the whole of Δ. On their axis, the limit they place at order six is already reached at order two: the bigram estimate has not converged on a corpus of this size.

The denominator is itself contested. Rao et al. [2010] normalise with $L = 417$ (Mahadevan). Parpola gives 386; Wells [2015]'s own published count is about 676, and the Interactive Corpus of Indus Texts (ICIT), the online sign list he maintains separately, currently sits at about 702. Changing only the denominator, corpus untouched, moves a value near 0.5 by about 0.045, the size of the whole of Δ in rongorongo, and that is a lower bound, since a change of sign list changes the string as well.

5 Size, not composition: a sweep over catalogue length

Section 4 showed that changing the catalogue moves the number. This section asks why. Two very different things could be going on, and only one of them is really about the script: a

	numeric		horley		freq		random	
L	Δ	cov	Δ	cov	Δ	cov	Δ	cov
25	−0.010	87 %	−0.025	94 %	−0.029	85 %	−0.016	100 %
50	−0.016	57 %	−0.045	66 %	−0.041	63 %	−0.030	83 %
75	−0.022	40 %	−0.060	45 %	−0.050	46 %	−0.039	59 %
100	−0.024	28 %	−0.068	31 %	−0.058	34 %	−0.045	41 %
125	−0.030	21 %	**−0.077**	24 %	−0.061	26 %	−0.049	31 %
200	−0.036	11 %	—	—	−0.064	14 %	−0.054	14 %
300	−0.042	6 %	—	—	−0.062	7 %	−0.055	7 %
633	−0.055	1.7 %	—	—	−0.055	1.7 %	−0.055	1.7 %

Table 2: Distance from the null (Δ) and bigram coverage as a function of catalogue size, four merge rules, same corpus. Full grid (15 sizes, z, $\hat{H}_c$) in the supplement.

longer catalogue could simply be too fine for what a corpus this size can populate, whatever it groups together; or two catalogues of the very same size could give different answers depending on *which* signs they group. Korovina (personal communication, 2026) named exactly this fork, "two very different things from a theoretical standpoint," and the sweep below is designed to tell them apart.

Three things follow from Table 2 and Figure 1(a).

There is no knee at the syllable count. Under the numeric rule the distance from the shuffle grows monotonically with L; under the frequency rule it rises to an interior maximum at $L = 200$ (-0.064) and declines slightly to the full inventory; under Horley's rule the largest value in his column is his published 125 (-0.077, $z \approx -77$ against the sweep's hundred permutations, -67 against the two hundred of Table 1; Δ is the same and only the null's spread differs), and merging Horley downward loses Δ steadily (-0.045 at 50, -0.025 at 25). No rule turns near the syllable count, and the numeric and frequency rules disagree about the *direction* of the effect above $L = 200$, which is why the merge rule has to be named wherever such a curve is quoted. Coverage, going the other way, falls below a quarter of the table once L passes 100–125. If "best" means *most structure visible while the statistic is still estimable*, the two criteria cross at roughly 100–125, where the field's own catalogue sits, and not at 50.

Size dominates. At $L = 633$ every rule is the identity and the four columns coincide; as L shrinks, all four lose Δ together, and the spread between rules at any fixed L (0.02–0.03) is smaller than the movement along L (0.05–0.06 between 25 and 633 for the same rule). Korovina's first hypothesis is the larger effect.

But composition is not zero, and its sign is informative. At every $L \leq 125$ the order is horley > freq > random > numeric. Horley beats random partitions of the same size by 5–11 random-draw standard deviations; frequency by 4–7; and the numeric rule, adjacent Barthel numbers merged, is *worse than random* by 3–10. Barthel's numbering follows shape, so this says that shape-adjacent signs behave differently in sequence, and merging them destroys structure that a random merge preserves by accident. §6 pursues this.

6 Composition: what the published catalogue knows

The previous section found that most of the movement is simply catalogue *size*: how many bins, not which signs land in which bin. This section asks about what's left over. For the entropy statistic, holding size fixed and changing only which signs are grouped together barely matters. But there is a sharper question than "does this change the number": does a catalogue capture graphic knowledge a mechanical rule would not guess? Checked against the scribes' own behaviour, the substitutions they actually made copying the same passage twice, composition turns out to matter a great deal.

At fixed L, holding the core. Korovina describes the field as agreeing on some 40–50 frequent signs and disagreeing about the rest. Holding the 50 most frequent numeric codes distinct (64.4 % of tokens) and cutting the 583-sign tail into 75 classes by three defensible rules (numeric adjacency, frequency, leading-digit family) at exactly $L = 125$ gives $\hat{H}_c$ of 0.758, 0.754 and 0.756; 300 arbitrary regroupings of the same tail give 0.7632 ± 0.0008. Composition alone, core held, moves the statistic by 0.004, one per cent of the length effect, though all three defensible rules sit below the arbitrary ones by $z = -6.6$ to -10.9, so the merge rule carries information.

At fixed L, rule free to touch the core. When the rule may merge frequent signs (§5), the spread across rules at fixed L is 0.06 at 125 and 0.14 at 50. Both numbers stand; they answer different questions. The first says that if researchers agree on the frequent core, their disagreement about the tail barely registers in conditional entropy. The second says that if they do not, it does.

Against the scribes. Conditional entropy is one question to ask of a catalogue. Another is whether it groups the signs the scribes themselves interchanged. Where the same passage recurs on different objects with a one-sign difference, the two signs at that position are candidates for allographs, the evidence Pozdniakov used by hand in 1996. The alignment runs in two steps (`parallels.py`), and the first only chooses what to compare. An index of every exact five-sign run selects candidate line pairs: two lines qualify if they share at least one such run and lie on different objects, a chance five-run over 633 types being negligible. Each candidate pair is then aligned in full (Ratcliff–Obershelp matching, Python's `difflib`, over the whole of both lines rather than by extension outward from the shared run) and kept only if some matched block is at least four signs long. Every position at which that alignment replaces exactly one sign by exactly one other is recorded as a substitution. This yields 119 cross-object passage pairs, 164 distinct substitution pairs, 266 events; the frequent ones, (002,021), (400,600), (045,046), (056,084) and (254,256), are numerically adjacent, which given Barthel's shape ordering is independent evidence that the alignments find variants rather than noise. Scoring catalogues on how often they place both members of a substitution in one class, all at $L = 125$, on the same 139 events the Horley map can classify:

Horley (published)	40.3 %
numeric proximity	18.0 %
leading-digit family	10.1 %
frequency	0.7 %
random, 300 draws	0.6 ± 1.1 %

The restriction from 266 events to 139 is forced rather than chosen: the Horley map returns nothing for the remaining codes, so there is no class in which to place them and no rule can be scored on them under a common test set. It is not neutral, and the direction is worth naming. The events dropped are exactly those the winning catalogue cannot classify, and on those it would have scored nothing; 40.3 % is therefore an upper bound on Horley's edge rather than an estimate of it. The three mechanical rules are scored on the same 139 events, so the ordering among them is unaffected.

Horley sits 35 standard deviations above chance and 2.2 times the best mechanical rule of the same size (a first pass that scored Horley on 139 events and the mechanical rules on 266 gave 2.9×; the common set is the fair one). Composition, then, is the *whole* effect where the scribes are the witnesses, and one per cent of it where conditional entropy is. The published catalogue encodes graphic judgement that no mechanical rule recovers; it simply does not surface in the statistic the debate has used. Limits: substitution pairs are candidates, not established allographs (two different words can differ by one sign); seeds require an exact match, so only

reference	L	reference ($N = 14\,812$ windows) $\hat{H}_c$	Δ	cov	rongorongo at same L $\hat{H}_c$ range	Δ range
Māori, length folded	52	0.684	−0.151	44 %	0.70–0.88	−0.015 to −0.046
Rapa Nui, length folded	55	0.695	−0.170	44 %	0.71–0.87	−0.017 to −0.047
Linear B	95	0.668	−0.156	26 %	0.71–0.79	−0.026 to −0.063
Māori, length kept	98	0.598	−0.177	21 %	0.70–0.79	−0.024 to −0.065
Rapa Nui, length kept	105	0.598	−0.187	19 %	0.70–0.78	−0.026 to −0.068

Table 3: Real syllabaries versus rongorongo merged to exactly the same L under the four rules of §5. Reference values are means over 20 windows of the numeric reading's token count, $N = 14\,812$ (s.d. of Δ 0.007–0.012). Rongorongo ranges span numeric, horley, freq and random; the freq rule gives 0.707 at $L = 55$.

variation in otherwise identical context is visible; 60 % of substitutions are not absorbed by Horley and we cannot tell whether the catalogue is still too fine or those passages differ in wording; and we find 119 cross-object pairs where Horley catalogued 146 passages by hand: the right order, not the same number.

7 Does a small catalogue make rongorongo look syllabic?

Here is the question a sceptic asks first: shrink rongorongo's catalogue down to about fifty signs, close to the number of syllables spoken Rapa Nui actually has, and does the script simply start to *look* like a syllable-based writing system, just because the numbers now land in the same place? Korovina (p.c.) predicted yes, and for a specific, honest reason: reducing the catalogue to about 50 "will inevitably get more similarity to syllables than to anything else," a property of the material rather than a bug in the test. Testing this needs syllabaries measured the same way, at the same L and the same N. Table 3 gives them; Figure 1 places them on the sweep.

The prediction is right on the axis the Rao–Sproat exchange is conducted on. At $L = 55$ the frequency-merged rongorongo comes out at 0.707 and syllabified Rapa Nui at 0.695: indistinguishable. Under the other rules the same corpus lands anywhere from 0.73 to 0.87. So at fifty the number can sit on the syllabary value or well away from it, and which happens depends only on which fifty. That is the "property of the material" Korovina describes, with a figure attached, and it is a second way of saying what §4 said: the level of $\hat{H}_c$ is not a property of the corpus.

The distance from the shuffle separates them. Every real syllabary, Rapa Nui, Māori and Linear B, sits 0.15 to 0.19 below its own shuffle at matched L and N. Rongorongo, under every catalogue from 52 to 105 and every rule, sits 0.02 to 0.07 below its own; and its bigram table is fuller (51–78 % of cells observed against 44 %): the signs combine more freely than syllables in a language do. A fifty-sign rongorongo can wear the syllabary's number, but it has less sequential structure than syllabic text of the same alphabet and length: 2.4–3.6 times less against the best rongorongo catalogue at each size (Linear B 2.4, Rapa Nui with length 2.8, Māori 3.3, Rapa Nui folded 3.6), and up to ten times less against the other rules. Linear B is the figure to put in front of a sceptic: an administrative script of lists and tallies, and it still comes out at −0.152 against rongorongo's best −0.064 at the same L.

The comparison above could still be an artefact of something that has nothing to do with the script itself: the wrong run length, too small a sample, one unrepresentative edition, the wrong genre, or the wrong reference language entirely. Each is checked in turn below, one at a time.

Two checks a referee would ask for. *Run length.* The rongorongo chain is cut by lacunae and line ends into 1 182 runs of mean 12.5 tokens; the references were cut at verse and inscription boundaries, and the Polynesian ones are longer (mean 28–36). Re-cutting each reference window into exactly the rongorongo run-length multiset before measuring moves the Polynesian values of Δ by less than 0.01 (Rapa Nui $-0.175 \to -0.167$, Māori $-0.152 \to -0.146$). Linear B runs are *shorter* (mean 3.9), so the legitimate direction is the reverse: cutting rongorongo into Linear B's run lengths moves its Δ from -0.064 to -0.062. (Cutting Linear B into rongorongo's longer runs would splice inscriptions together and invent adjacencies; it lowers the Linear B figure to -0.107, and we do not use it.) *Sample size.* Measured at 25, 50, 75 and 100 % of the rongorongo token count with run lengths matched at each N, every series is flat: rongorongo Horley-125 goes -0.066, -0.070, -0.075, -0.077; Horley merged to 55 stays between -0.045 and -0.048; the references stay within their window-to-window scatter. The gap is not a sample-size artefact and would not close with more tablets; rongorongo is already near its asymptote at a quarter of the corpus.

Native prose, and what the four collections actually are. The references above are Bible translations, which is what was reachable. Korovina (p.c., 22 August 2026) supplied four collections of Rapa Nui text and, on being asked, described them: the Isla6 edition is school textbook material in the modern language; Englert's texts and Manuscript E are legend narratives of the 1920s and 1930s or a little earlier; and Campbell's corpus is song lyric, some of it in Tahitian rather than Rapa Nui. Together they give 167 077 syllable tokens after Spanish and editorial material is removed at word level, each removal cutting the chain (`korovina_ref.py`). The three narrative collections confirm the gap: at matched L and N, Englert gives -0.157, Isla6 -0.156, Manuscript E -0.154 and the pooled corpus -0.145, against rongorongo's best of -0.045 to -0.066 at the same sizes. Native prose gives a slightly *smaller* Δ than the Bible, the translation being the more formulaic text, so the earlier reference was if anything flattering to the language side.

The fourth collection dissents, and the reasons are measurable. Campbell gives -0.083. Three quarters of the difference are edition and sample rather than language; the rest is genre. It is the only one of the four marking neither vowel length nor the glottal stop (glottals appear 3.0 times per thousand characters against Isla6's 46.1), and the glottal is a phoneme, so leaving it unwritten merges syllables the language distinguishes. Degrading Isla6, the best-marked collection, one respect at a time and remeasuring (`campbell_probe.py`): stripping its glottals and macrons moves its Δ from -0.165 to -0.130, 43 % of the distance to Campbell; truncating it to Campbell's token count moves it 26 %; cutting its chain at Campbell's word-drop rate moves it 5 %. The remainder is accounted for by what Korovina told us afterwards, that the collection is song and partly a different language. It is the argument of §4 arriving where it was not expected: two editions of one living language, measured the same way throughout, differ by a factor of two, and most of the difference is in the transcription.

Does the comparison hold against the right genre? The objection this invites is that the tradition names chanted genres, *kohau ika*, litanies and genealogies, while every reference here is prose. Hawaiian answers it, being the one Polynesian language with a large song corpus and a large prose corpus surviving from the same culture: 1 329 song texts from `huapala.org` against Fornander's collection, 164 143 syllable tokens each, matched run lengths, and both normalised to unmarked orthography because the song texts mark the glottal and Fornander does not (`hawaiian_genre.py`). Song and prose differ, but not much and not in the direction that would rescue rongorongo: Δ -0.0885 against -0.0804, *ABAB* 7.20 against 4.33, *AA* 0.46 against 0.59. The difference between the two orthographies of the *same* songs is larger than

reference	$L = 30$	$L = 40$	$L = 55$	$L = 90$
Rapa Nui, New Testament	−0.121	−0.148	−0.149	−0.149
Rapa Nui, Isla6	−0.114	−0.135	−0.132	−0.132
Māori, New Testament	−0.115	−0.125	−0.123	−0.123
Linear B	−0.083	−0.106	−0.118	−0.110
Hawaiian prose	−0.093	−0.102	−0.102	−0.102
Hawaiian song	−0.073	−0.076	−0.076	−0.077
rongorongo, best rule	−0.022	−0.027	−0.034	−0.049

Table 4: Every reference merged to the same catalogue size by frequency, at the token count and run lengths of the Horley reading on the objects of at least 100 glyphs: $N = 15\,115$ in 1 681 runs, not the 14 812 of Table 3 (`crosslang_L.py`). The rongorongo row is the largest $|\Delta|$ among the rules available at that L (numeric, Horley where $L \leq 125$, frequency), taken column by column rather than fixed in advance. Maximising favours rongorongo, so the ratios here are the smallest this comparison can be made to yield. The references saturate above $L = 40$; rongorongo does not, which is the sweep of §5 seen from the side. Ratios run from 1.55 to about 5.5.

the difference between the genres: Δ −0.1089 marked against −0.0885 unmarked, 23 % against 10 %.

Genre moves the level; sample size moves the shape; neither moves the conclusion. Four Māori genres measured on one language remove the remaining ambiguity (`genres_mri2.py`). Chant (Grey's 1853 *mōteatea*, 111 432 words) and gospel narrative peak at the same catalogue size as each other at every sample size tried, and that size moves with N rather than with genre: $L = 55$ at 5 000 and 12 000 words, $L = 90$ at 25 000. Rongorongo's peak at $L = 90$ with $N = 15\,115$ (the 15-object subset of Appendix A) is therefore consistent with word-level language at comparable size and distinguishes nothing. What genre does move is the level, and substantially: narrative carries 1.7 times the structure of chant at matched size, because chant's vocabulary is far richer: 3 235 word types against 1 380 in the same number of words. Two genres are too small here to place a peak and are reported as such: Māori teaching at 1 216 words and genealogy at 339.

Both size and reference language move the ratio. Table 4 puts every reference at the same L, N and run lengths. The gap runs from 1.55 (Hawaiian song at $L = 90$, the weakest reference at the size most favourable to rongorongo) to about 5.5 for the Rapa Nui New Testament at $L = 30$. A single figure of 2.4–3.6 sits inside that range, but it is one size against Polynesian prose. Size and choice of reference each move the answer about twofold. Saying so is more in keeping with this paper than defending a single figure would be.

Name lists do not match either. Since the tradition names genealogies, they are the genre most likely to resemble the script. At word level, against a control of the same size, they do not: pooled Rapa Nui and Māori genealogy (Matthew 1 and Luke 3, 3 533 words) sits highest at $L = 20$ and declines, where size-matched prose rises to $L = 90$ (`genealogy.py`). A genealogy's structure lives in its formula, *A X he poki 'a Y*, which is built from the commonest words, so a small catalogue already holds it and rare names dilute it. This cuts against the list reading, and against our own earlier leaning towards it.

We do not draw a conclusion about the script from this. It is consistent with rongorongo not being a plain syllabary of Rapa Nui, and also with a syllabary heavily obscured by ligature, allography and layout conventions our merge rules do not model. What it rules out is that the resemblance at $L \approx 50$ on the raw axis is evidence of syllabic structure: the axis cannot tell the

	rongorongo (CEIPP, Horley)	Indus (Mahadevan 1977)
tokens / mean run	14 812 / 12.5	13 372 / 3.7
best Δ (z)	−0.077 (−67)	−0.217 (−169)
vs. syllabified Rapa Nui, same N, run lengths, L	2.4–3.6× less	0.7× (more)
AA vs. within-text shuffle	1.12×	0.9×
$\hat{H}_c$ span, full size sweep	0.318 (125–1 897) = 4.1× Δ	0.384 (25–417) = 1.8×
$\hat{H}_c$ span, published catalogues	catalogues 50–600 (10×)	catalogues 386–702 (1.8×); 300–417 measured: 0.16× Δ
sweep shape (freq. rule)	interior optimum $L = 200$, shallow	interior optimum $L \approx 150$–200
shape-ordered merge vs random	worse than random (−3 to −10 sd)	equal (−1 sd)

Table 5: The same statistic and the same null on two undeciphered corpora. Code and results: `indus/`.

two apart, and the axis that can does not show it. Limits: the references now span narrative, song and name list, and disagree with one another by up to a factor of two at matched size; the (C)V syllabification is mechanical and does not separate diphthongs; Linear B is the only reference that is itself an ancient inscribed corpus, which is why we lean on it.

8 Noise, sample and estimator: real, and an order of magnitude smaller

One more thing has to be ruled out before any of the above can be trusted: could the whole effect just be noise: typos in the transliteration, too few tablets, or a biased way of estimating entropy from thin data? Each is checked below. Each turns out to be real, and each is an order of magnitude too small to explain what Sections 4–7 found.

Transliteration error. Korovina notes that all existing transliterations are “dirty” in the sense of containing errors rather than merely recodings, “although the error rate is not very high.” Replacing a fraction p of glyph tokens by another sign drawn from the corpus unigram distribution (20 draws each) moves $\hat{H}_c$ by +0.002, +0.006, +0.010, +0.020 at $p = 1, 3, 5, 10\,\%$ under the numeric reading and +0.003, +0.008, +0.013, +0.024 under Horley’s; Δ erodes by the same amounts. Five per cent error is about a thirtieth of the catalogue-length effect and a sixth of Δ; ten per cent removes a third of it and leaves the rest. It cannot account for the spread in Table 1, nor for the gap in Table 3. The error model is simple, in that errors do not preferentially fall on look-alike signs, and is offered as an order of magnitude.

Estimator. The plug-in estimator is biased downward on sparse tables. Miller–Madow correction, applied to unigram and bigram before subtraction and to real and null alike, changes the ratio of §4 from 4.1 to 3.7, does not reorder the readings, and barely shrinks the spread (0.318 → 0.309); Δ grows slightly (0.077 → 0.083) because the null moves more than the data. But Miller–Madow assumes $m/N \to 0$, and here the number of observed bigram types over bigram tokens is 0.67, 0.51 and 0.27 for the three readings: most bigrams are seen once. In that regime no plug-in-family estimator is reliable, and corrected levels should not be quoted as absolute either. *Differences*, between readings and between data and null, survive because the bias is on both sides.

L	Δ	coverage	
150	-0.251	9.7 %	maximum of the sweep
300	-0.229	3.0 %	
386	-0.220	1.8 %	Parpola
417	-0.217	1.6 %	Mahadevan, full inventory

Table 6: Indus, frequency-merged to each size, pooled-corpus null as in Table 5 (`indus_published_range.py`). Parpola to Mahadevan moves Δ by 1.5 %; the coverage column is the qualification.

L	rongorongo	Indus	ratio	
125	26.2 %	12.7 %	0.49	Horley, the field's own catalogue
150	20.6 %	9.7 %	0.47	Indus maximum under this rule
200	13.6 %	6.2 %	0.46	rongorongo maximum under this rule
386	4.5 %	1.9 %	0.41	Parpola
417	3.9 %	**1.6 %**	0.41	Mahadevan, full inventory
633	**1.7 %**	—	—	CEIPP numeric, full inventory

Table 7: Share of the $L \times L$ bigram table observed, both scripts on one grid (`coverage_symmetry.py`): same statistic, same merge rule (frequency fold, keeping the $L-1$ commonest signs), same sizes, and no bigram crossing a text boundary. 14 812 rongorongo tokens yield 13 630 bigrams; 13 372 Indus tokens yield 9 798, because Indus texts average 3.7 signs. Coverage is what §5 uses to bound rongorongo's usable catalogue range, and it is applied here to the script the paper was in danger of exempting from it.

9 Robustness: one test applied to every claim

A paper arguing that a measurement choice moves the numbers has to answer the same question about its own numbers. The natural test is the one that matters most for a corpus of twenty-five objects: change the null from a shuffle of the pooled corpus to a shuffle within each object, so that rongorongo is measured the way a single-text reference is measured; treat the two codes that are not signs as breaks in every reading (§2); and recompute the parallel-passage seeds without the unidentified-glyph code. Every quantitative claim in the paper is then remeasured under all three at once (`robust_all.py`). Table 8 gives the result.

Four of the five hold. The ratio in Table 1 and the syllabary gap both move *up* under the stricter treatment; Horley's edge over the best mechanical rule moves *down*, from 2.2× to 2.1×, still comfortably more than double, so composition holds by a smaller margin rather than a larger one. The Indus row is a different kind of comparison: it has no pooled-null figure to move from, since it is itself the ratio between two nulls. The fifth claim does not hold, and the reason is instructive rather than embarrassing. Conditional entropy is a property of the whole bigram distribution: it averages over every cell, and a tenth of Δ is the most the null can take from it. The immediate-repetition rate is a single rare event type, and against a pooled null that mixes the vocabularies of twenty-five objects it reads as two and a half times what it is. The measurement that fails is the one most exposed to vocabulary mixture; the measurements that hold are the ones that average over it. The paper's central statistic is robust to a choice that one of its appendices is not, and the appendix says so in its own voice.

One further check is reported because it would have misled us had we not run it to the end. Rewriting the parallel-passage search from scratch, instead of rerunning the original script, produced a Horley absorption of 5.8 % against the 40.3 % of §6. The new implementation was the one at fault: it recorded every mismatch within forty positions of a seed, and so found 1 609 putative substitution pairs where the careful alignment finds 164, most of them noise.

	claim	pooled null	object null	
§4	catalogue spread, × distance from null	4.14	4.28 / 4.60	holds, larger
§6	Horley against the best mechanical rule of its size	2.2	2.1	holds, smaller
§7	gap to real syllabaries, best catalogue	2.4–3.6	2.5–3.9	holds, larger
§10(4)	Indus Δ retained under a within-text null	—	72 %	holds
App. A	rongorongo AA, times expectation	1.6–2.4	1.12	**does not hold**

Table 8: Every quantitative claim remeasured with the two non-sign codes removed and an object-preserving null. For §4 the two figures are with the codes removed alone, and with the null changed as well. For §10(4), Indus keeps 72 % of its pooled Δ under a within-text shuffle where rongorongo keeps 71–75 %, so the comparison between the scripts is unchanged. AA: the rate of immediate sign repetition, one sign following itself; defined and measured against language in Appendix A.

The original reproduces exactly. A rough new implementation that appears to overturn a careful old one is a trap a reader of this paper should know the shape of, because it is the same trap as the catalogues: a different procedure producing a different number, and the difference read as a finding.

10 What follows

Five things the measurements above support, and, where it matters, the line past which they do not.

1. *Levels do not travel; differences do.* A normalised conditional entropy of 0.4 or 0.7 for an undeciphered script is a statement about the catalogue and the sample before it is a statement about the script. What can be compared across corpora is the corpus's distance from its own unigram-matched null, and even that only at matched L and N.

2. *Report the catalogue as a parameter.* Any statistic on an undeciphered corpus should be reported as a curve over catalogue size, or at least at two published granularities, with bigram coverage alongside. A single number at a single L invites the reader to compare it with a language plotted at another L, and §7 shows how misleading that comparison can be.

3. *The field's catalogue is where the statistic peaks.* The ∼125-sign inventory reached by hand is where structure is maximal and the table still a quarter populated. That is a small vindication of the epigraphers, and a reason not to expect statistics to improve on their judgement about *size*; what statistics can add is a test of *composition*, and there the scribes' substitutions, not conditional entropy, are the evidence.

4. *The same statistic gives the opposite verdict on Indus, which is the point.* If the measurement itself were the problem, it should be unstable on every script it touches. So we ran the identical code on a completely different undeciphered writing system, the Indus corpus: Mahadevan's 1977 concordance as digitised by Hamilchin [2026], 3 574 text lines and 13 372 sign tokens over 417 signs, uncertain readings and lost signs breaking the chain; and the 1 548-text Extended Basic Unique Data Set (EBUDS) of Rao et al. [2009], their filtered version of Mahadevan's concordance with damaged, illegible and duplicate texts removed, whose normalised bigram figure we reproduce at 0.387. Table 5 sets it beside rongorongo. Indus texts are short, 3.7 signs on average, and rigidly positional (the terminal "jar" sign 342 is a tenth of all tokens), and the corpus sits 0.217 below its shuffle ($z \approx -169$), *further* than syllabified Rapa Nui, Māori or Linear B at the same N and run

lengths. It avoids immediate repetition where rongorongo, under the same within-text null, does not (*AA* 0.9 against 1.12; Appendix A).

It is not only Δ that differs between the two scripts. The catalogues themselves behave differently too, and this is what makes Indus a genuine outside check rather than a coincidence. Sweeping size from 25 to 417 moves $\hat{H}_c$ by 0.384, 1.8 times Δ; but the published Indus inventories disagree only 1.8-fold (Parpola's 386, Mahadevan's 417, and Wells [2015]'s own count and his separately maintained ICIT list at about 676 and 702), and the curve rises to a maximum near $L = 150$ and then declines, so that every published inventory sits on its far slope. Between the two we can reach by merging, the statistic moves by 0.003, 1.5 % of Δ (Table 6). Wells's granularity cannot be reached: merging only goes down, and the one digitised Parpola coding we can find is 179 Mohenjo-daro artefacts. Keeping Mahadevan's uncertain readings as their own types (561 against 417) moves the statistic by 0.02, and Mahadevan's numbering, unlike Barthel's, carries no sequential information: merging by it equals random within one standard deviation.

So the catalogue problem is script-specific, and a question from R. Sproat (p.c.) made us say precisely how. On Indus the distance from the null is large and the inventories close; the statistic is stable against the choice of catalogue over the range we can measure, and untested above it. It is *not* stable against estimation at those sizes, and saying so properly means applying §5's coverage criterion to both scripts on one grid rather than asserting it of Indus alone. A criterion applied to one side only is the asymmetric comparison this paper is about. Table 7 measures it: one statistic, one merge rule, the same sizes, no bigram crossing a text boundary. Indus coverage is 0.41 to 0.54 of rongorongo's at every matched size and the ratio moves smoothly, so it is a property of the two corpora rather than of one catalogue: the token counts differ by 10 % (13 372 against 14 812) but the bigrams they yield differ by 28 % (9 798 against 13 630), because Indus texts are short and every text boundary costs a bigram. At the inventories people actually publish the consequence is blunt. Mahadevan's 417 signs leave 1.6 % of the Indus bigram table observed, slightly less than the 1.7 % of rongorongo's 633-sign numeric reading, the reading §4 places on the unestimable side of its coverage column and §5 puts outside the usable range. Applied to both, that criterion gives $L \leq 125$ for rongorongo and $L \leq 50$ for Indus at 25 % coverage, $L \leq 200$ and $L \leq 125$ at 10 %; the published Indus catalogues sit eight times past the stricter line. What keeps this from being an accusation is the other half of the same table. Under the frequency rule, applied to both, the two curves have the same shape, an interior maximum and a decline: rongorongo peaks at $L = 200$ (-0.064) and falls to -0.055 at 633, Indus peaks at $L = 150$ (-0.251) and falls to -0.217 at 417. Each published catalogue therefore returns most of its own maximum: Horley's 125 gives 95 % of it at 26 % coverage, Mahadevan's 417 gives 86 % of it at 1.6 %. The statistic is stable in both scripts; only its estimability differs, by a factor of sixteen. Nair [2026], working at one ICIT granularity, writes that "no fully validated cross-concordance exists" and that his 584 "should not be directly compared" with 417 or 386; the figures above say what that caution is worth in each direction. What Indus's large Δ *means* is a separate question (it is positional structure in very short texts, which Sproat [2014] argues non-linguistic systems show as well), and we do not enter it.

11 Conclusion

In plain terms, this is what the measurements say.

Rongorongo has no agreed list of signs. The published lists differ by more than a factor of ten, and every statistic anyone computes about the script is computed after choosing one of them. We took the same carvings, the same twenty-five objects, and changed nothing except which marks are counted as the same sign. The headline statistic moved about four times further than the thing it is supposed to be detecting. So when a paper reports that rongorongo scores 0.4, or 0.7, on a measure of how predictable one sign is from the last, most of what that number records is the transliteration, not the script. Numbers of this kind cannot be carried from one study to another and compared, because they are not measurements of the same object.

That does not make the statistic useless. What survives being moved from one list of signs to another is not the level of the number but the distance between the real corpus and a shuffled copy of itself, and that distance can be compared, at a matched inventory size and a matched amount of text, and not otherwise. Measured that way, rongorongo carries real but modest sequential structure, less than a working syllabary of a real language does at the same size. We do not conclude from this that the script is not a syllabary. The comparison is not sharp enough to decide it, and we say where it stops.

Two results point the other way, in the epigraphers' favour. The list of about 125 signs that specialists reached by hand, without statistics, turns out to sit almost exactly where the structure in the corpus is greatest and the data still sufficient to estimate it; a mechanical rule of the same size does markedly worse at grouping the variants that scribes themselves treated as interchangeable when they copied a passage twice. And the pattern Davletshin identified (two signs alternating, *ABAB*, as Polynesian languages double a word for emphasis or plural) is present across the whole inventory at the rate Rapa Nui and Māori show when they are written syllable by syllable, and is absent from a syllabary of a language that does not reduplicate. What we could not find is his division of the signs into two classes; the pattern is spread over the inventory rather than concentrated, with his *ki* the one clear individual exception.

The last result is the one we did not go looking for. Running the identical code on the Indus script returns the opposite verdict: there the published sign lists agree closely and the corpus sits far from its shuffle. So the problem measured here is a fact about rongorongo's particular transliteration difficulty, not a general defect of statistics applied to undeciphered writing. The practical recommendation is the one in §10: report the sign list as a parameter, not as a settled fact, and report how much of the table your data actually fills.

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A A calibrated decipherment attempt

A fair test of a lock needs a key you already know doesn't fit, so you can see what "doesn't fit" looks like before trusting a verdict on the lock that matters. That is what *calibrated* means below: before asking whether rongorongo maps onto Rapa Nui syllables, we first check whether the same method would also, wrongly, claim success on material we know is not Rapa Nui. The natural objection to §7 is that a bare conditional entropy might miss a syllabic mapping that a decipherer would find, so we ran the obvious one and calibrated it first. The model maps the 125 Horley glyphs many-to-one onto the 55 Rapa Nui syllables, scored by an add-k bigram model of syllabified Rapa Nui (247k syllables, 20 % held out) plus a channel term $\sum_g n_g \log(n_g/N_{m(g)})$ that charges for homophony; without that term the search collapses everything onto two vowels and "beats" the true mapping. The search, a standard randomised algorithm that proposes many candidate glyph-to-syllable mappings and gradually settles on likely ones (Metropolis with annealing), runs 300k proposals and six restarts; how much the six restarts agree on a given glyph's mapping is read as how confident the search is about it, everything at the rongorongo token count and run lengths.

Calibration first. Held-out Rapa Nui enciphered by a random one-to-one substitution is recovered in full: every token, with two of six restarts reaching the true-mapping likelihood. A homophonic cipher with 99 symbols is recovered to 37 % of tokens, with a likelihood gain of +0.35 nats per token over its own shuffle. Then the script. Rongorongo under its best mapping gains +0.17 over its shuffle ($z \approx 21$), and Linear B, which is Greek, pushed through the same pipe gains +0.15; Māori, a sister language, +0.27. The decipherer reads rongorongo as Rapa Nui exactly as well as it reads Linear B as Rapa Nui. It is finding sequential structure, not language. No glyph locks across restarts: among the 25 most frequent, the best any of them does is agreement on three of six restarts, true of a dozen of them including Horley 76, and the rest sit lower, at one or two of six. This does not exclude a logographic-plus-syllabic mixture (some glyphs standing for whole words, *logograms*, mixed with others standing for syllables), ligatures as words, or an anchored attack; it does say that a plain one-glyph-one-syllable hypothesis under a bigram model has no more support from the corpus than an unrelated script would give it (`decipher.py`).

We also tried the one anchor the corpus offers, the lunar calendar of Mamari (Ca6–9; Guy, 1990). The text has 28 crescents (Barthel 040) in groups of 2/6/3/7/3/5/2 after an initial group with no crescents at all, a heralding sequence repeated eight times, and the full-moon glyph 152 near the middle, as Guy describes. Clamping glyph 040 to each of the 55 syllables in turn and re-running the global search ranks the candidates as noise, with *ma* (mahina, marama) sixth, *hi* nineteenth and *po* thirtieth of 55 and a total spread of 0.11 nats per token, and the herald reads, under the best mapping, as a string the language model likes no better than chance ($z = 1.7$). The calendar supports the crescent as a pictogram, not a syllable. It also supplies one more instance of the paper's point: the Horley map folds 040 and 041 into one class and does not reach 152 at all, so the distinction Guy reads, night crescents against herald crescents and the full moon, is invisible under the very catalogue on which the statistics are computed (`mamari.py`).

A mixed model, each glyph either a syllable or one of 203 frequent Rapa Nui words with a logogram prior, was built as well, and it *failed its own control*: on held-out Rapa Nui with thirty frequent words replaced by logogram symbols it recovers at most 4.5 % of tokens at any prior. Its verdict on rongorongo therefore carries no weight either way: the same pattern, gain over shuffle 0.10 against Linear B's 0.11 and the control's 0.14, with moon and night words for the crescent all scoring below the free choice. We report it as a limit of the method at this corpus size, not as a finding about the script (`mixed.py`).

	shuffle pooled corpus			shuffle within each text			shuffle within each run		
	AA	*AAA*	*ABAB*	*AA*	*AAA*	*ABAB*	*AA*	*AAA*	*ABAB*
rongorongo	1.59	4.34	34.6	**1.12**	**1.68**	**20.7**	0.62	0.63	4.7
Rapa Nui (NT)	0.57	0.47	8.3	0.57	0.41	9.3	0.57	0.55	8.1
Rapa Nui (Isla6)	0.69	0.29	7.7	0.69	0.31	8.0	0.65	0.25	5.8
Māori (NT)	0.52	0.19	4.8	0.52	0.16	4.7	0.52	0.20	3.5
Linear B	0.72	4.38	1.2	0.74	4.33	1.1	0.64	2.56	0.8

Table 9: Observed repetition divided by its expectation under three nulls (`null_choice3.py`). References are re-cut to the rongorongo run-length multiset before measuring. The middle block is the null that matches how the languages were measured: one shuffle per text, one object being one text.

A repetition profile, and the null it depends on. Under any one-to-one key the rates of immediate repetition (*AA*), of triple repetition (*AAA*) and of period-two repetition (*ABAB*) in the glyph stream equal the rates in the syllable stream, so a repetition profile is inherited by every syllabic reading whatever the key. That makes it worth measuring, and it also makes the null decisive. Against a shuffle of the whole corpus pooled together, rongorongo shows *AA* at 1.6–2.4 times expectation, *AAA* at 5–9 and *ABAB* at 30–130, where syllabified Rapa Nui and Māori sit at *AA* 0.5–0.6. That contrast is real and it is also misleading, because the corpus is twenty-five objects with different vocabularies and its pooled unigram distribution is a mixture; any locally concentrated stream is repetition-seeking against a mixture by construction. The reference languages are each a single homogeneous text, where no such gap between pooled and local can arise. The two sides are only comparable under one shuffle per text, one object being one text. Table 9 gives all three nulls.

One consequence has to be stated before any count below is read. A shuffle that preserves objects is meaningless on an object holding a handful of glyphs, so objects under a token floor are dropped, and two floors are in use. At 100 tokens under the Horley reading, 15 of the 25 objects survive, with 1 681 runs and 15 115 tokens; this is the subset behind the sign-level counts below (`davletshin_classes.py`, `davletshin_locate.py`) and behind the matched-N comparisons of §7. At 300 tokens, 11 objects survive, and this is the subset behind Table 9, behind the per-object repetition profile, and behind the ligature and seed-stability checks whose figures are quoted below. The two are not interchangeable and neither is a corpus total: the same *ABAB* count is 201 events on the larger subset and 193 on the smaller, and the ligature share of *AA* is 18.8 % against 19.2 %. Both floors are reported side by side in `ligature_aa_results.txt`. Which is which below: the event counts (201 *ABAB*, 56 *AAA*) are the 100-token figures, and the ratios quoted against a null (*AA* 1.12 falling to 0.91, *ABAB* 21.0 falling to 17.6) are the 300-token ones.

The references do not care which null is used, syllabified Rapa Nui sitting at 0.57 on *AA* under all three, because each is one homogeneous text. Rongorongo cares a great deal. Under the matched null it sits at *AA* 1.12 against the languages' 0.52–0.74, at *AAA* 1.68, and at *ABAB* 20.7 against 4.7–9.3. One of those three does not survive the ligature check of Appendix A, and it is the one nearest the boundary: 19 % of the *AA* events are one glyph split into two identical components, and with them removed from observation and shuffle alike *AA* falls to 0.91 (`ligature_aa.py`). Rongorongo does not repeat a sign immediately more often than its own null predicts; it repeats *ligatures* whose halves are equal. *AAA* is barely touched, at 3.6 % ligature-borne. Two things follow. A syllabary can produce a profile of this kind: Linear B, an actual syllabary, sits at *AAA* 4.33, above rongorongo, so nothing here rules one out. And what is real is smaller than the pooled figures suggest: on *ABAB* rongorongo stands above every reference under every null, by a factor of about two.

The movement is worth more than what it was meant to prove. The languages are null-independent and rongorongo is not, and that is itself a measurement: rongorongo's repetition statistics are governed by local vocabulary concentration rather than by adjacency. Within a run of about twelve signs, shuffling that run's own tokens produces *more* immediate repetition than is observed, 813 expected against 496, while doing the same to twelve syllables of Rapa Nui changes nothing at all. The signs of a rongorongo passage are drawn from a much narrower local pool than the syllables of a Rapa Nui passage are, and the adjacencies within that pool are, if anything, avoided. That is stable under every null tried here, it is consistent with the list, tally and litany readings the epigraphers favour, and it is not evidence against a syllabary.

Which signs carry the repetition. Davletshin [2017] proposes that rongorongo signs fall into two combinatorial classes: those that form $ABAB$, $BABA$, $AAAA$ and AAA sequences, which he takes to be syllabic, and those that stand in isolation, which he takes to be word-signs; Davletshin [2022] reads the $ABAB$ sequences as the full reduplication that East Polynesian uses for intensives and plurals (*nui-nui*, *hatu-hatu*), and observes that triple repetition of a word makes sense in no language. That is a claim at the level of individual signs, and it can be counted (`davletshin_classes.py`). Under the Horley reading and the within-object null, the 201 $ABAB$ events of the 15-object subset fall on 74 of the 125 sign types, and the commonest tenth of types carries 57 % of them where frequency alone would predict 80 %: the pattern is spread across the inventory roughly in proportion to how often each sign occurs, not concentrated on a class. Thirty of those 201 events, and four more in part, exist only because the Horley map splits a ligature into its components: their alternating slots are not four glyphs on the object but two, one sign written twice. That is 17 % of the count, and it is a property of the reading rather than of the object (`davletshin_locate.py`). It came to light because Albert Davletshin, asked to comment on four events, asked in return for the lines they sit on rather than for the numbers. The two syllabic signs we can identify from his descriptions without guessing, "*Staff" (his *ki*, which is Barthel 001 and the commonest sign in the corpus) and "Sitting Man" (his *ka*, Barthel 380s), rank first and seventh among $ABAB$ carriers, and the ligature correction separates them.[2] Applied per sign rather than only to the total, it leaves 30 of *Staff's 31 events standing on four distinct glyphs and only 8 of Sitting Man's 17. Against a median corrected the same way (1.6 events per hundred tokens over the 69 types with at least 50 tokens, down from 2.4 uncorrected), *Staff sits at 3.9 per hundred, 2.4 times that median, and stays first of the 125; Sitting Man sits at 1.7, 1.04 times the median, and falls to twelfth. One of his two syllabic signs behaves differently from a sign of its frequency and the other does not. That is weaker than a class and stronger than nothing, and it is the reason the class test above and this one disagree: a single sign can stand well clear of the median without the inventory dividing in two.

The comparison that matters is with language. Syllabified Rapa Nui and Māori, cut to the rongorongo run lengths and token count, spread their $ABAB$ the same way, over 68 and 54 % of syllable types, at the same per-token rate. Linear B, a syllabary of a language without reduplication, gives eleven events in fifteen thousand signs, on seven types. And Rapa Nui written as words gives none at all, because *nuinui* is one word. None of the references can produce the ligature effect: their tokens are syllables cut from running text, one to a slot, so

[2] Corrected 2 September 2026, after the versions deposited to date. Those gave *Staff as Barthel 200, and the figures reported for it in this sentence were 200's: fourth of 125 at 2.5 events per hundred tokens, which sat on the corpus median and supported the opposite conclusion to the one below. Rafał Wieczorek supplied the right code (p.c., 2 September 2026), and three checks agree with him. Barthel 001 is drawn as a plain notched vertical bar where 200 is an anthropomorph with limbs. Davletshin [2022] calls *Staff the most frequent sign in the corpus with 775 examples, and 001 carries 787 tokens on the numeric reading against 313 for 200. And 200 appeared commonest here only because Horley's map folds 201–207, 210, 300, 462 and 548 into it, while nothing at all folds into 001. The error was itself an instance of this paper's subject, a property of the reading read as a property of the script. Corrected for ligatures, 200 sits at 1.4 events per hundred, 0.84 of the median.

the comparable quantity is the rongorongo count with the events that stand on fewer than four glyphs removed from the observed stream and from inside every shuffle alike. That restriction takes rongorongo's $ABAB$ from 21.0 times expectation to 17.6, a sixth of it, on the 11 objects of at least 300 tokens and on four hundred shuffles rather than the twelve behind Table 9 (`abab_null_stability.py`). Four hundred are needed: the expectation here is about nine events, and twelve-shuffle estimates of the same ratio range from 19.1 to 26.9 across seeds, so Table 9's 20.7 should be read as roughly twenty, not as 20.7. The margin over the references narrows and the verdict does not turn. $ABAB$ is therefore not a property of syllabic writing; it is the signature of Polynesian reduplication written syllable by syllable, and rongorongo carries it at the Polynesian rate across the whole inventory. That is stronger support for Davletshin's reading of the pattern than a separate class would have been, since it says the stream as a whole behaves like the syllables of a reduplicating language. AAA is the other half of his claim, and it goes the other way: 56 events in rongorongo (44 with the Mamari crescents removed) against 6 in Rapa Nui and 7 in Māori. The language does not do it and the script does. Two patterns, two verdicts, is what a logosyllabic system would give, with the triple repeats coming from whatever the non-phonetic part is.

One limit on all of this comes from Davletshin himself. Shown the six alternations above as lines rather than as counts, he replied that he holds every one of them in his own files and classes none of them as morphological reduplication: he reads them as *rhetorical* repetition, on the evidence of parallel passages and of other rhetorical repetition nearby (p.c., 1 September 2026). Doubling a word for emphasis and repeating a phrase for effect leave the same trace in a count of $ABAB$, and nothing measured here separates them. What the cross-language comparison establishes is narrower than the paragraph above may read: the *rate* at which rongorongo alternates matches syllabified Polynesian and not a syllabary of a non-reduplicating language, which is a fact about the rate and not about which of the two mechanisms produces it. Rapa Nui and Māori cut to these lengths contain rhetorical repetition of their own, so the reference is not clean in that respect either. The distinction is his, it is not one this measurement can make, and it is the reason we call $ABAB$ a signature of the pattern rather than evidence of a morphology.

The six logographs, checked. Davletshin [2022] names six logographic readings (MEA RED, TOKI ADZE, TURTLE, SHELL, SMALL SHELL, URCHIN) by description rather than by Barthel number; asked directly for the codes, he supplied Horley numbers for all six (p.c., 25 August 2026), one of them qualified: MEA RED at 11, which he suspects may conflate two distinct signs; TOKI ADZE at 63; TURTLE at 280; SHELL at 70; SMALL SHELL at 17; URCHIN at 102 (`davletshin_signs.py`). Seven frequent Horley signs (8, 66, 50, 71, 11, 522, 532) take part in no $ABAB$ event at all; of these, only MEA RED (11) is among his six. The other four large enough to test each take part in $ABAB$ at least once, though the raw window counts overstate by how much. Of TOKI ADZE's four events on 155 tokens, one is the ligature case: Db0301-01 and -02 are both written 383, one sign twice, and only the split makes an alternation of it; three survive. SHELL's three on 164 tokens are one alternating stretch on a single line, Ab0605-11 to -20, counted three times by a window that overlaps itself; one survives. TURTLE has one event on 119 tokens and SMALL SHELL one on 28. URCHIN, on 6 tokens, is too rare to test either way. Read as lines rather than as counts, none of the six surviving events is a logograph repeating itself: in each, one of the two alternating signs recurs through the whole line as a frame (63 on Er lines 1 and 2, 70 on Ab line 6, 53 on Er line 5, 1 on Db line 3 and Ev line 4), and the alternation registers where that frame's neighbour happens to occur twice. One of six lands on a zero-$ABAB$ sign, and even that match rests on an identification its own author is unsure of: weak evidence for his readings, not confirmation. It is reported in the direction it came out, as promised.

The Santiago Staff, measured rather than read. Fischer [1995] read the Staff as procreation triads X–76–Y–Z, noting himself that glyph 76 "is attached to the first glyph in each section". The reading has been criticised at length since: by Guy, by Pozdniakov and Pozdniakov [2007], and by Robinson, who counted only 63 of 113 sequences obeying the structure. Sproat [2026] removed 76 from the corpus and recomputed approximate string matches across objects, finding that the Staff remained an isolate.[3] The measurement here is a different quantity, repetition within each object against that object's own shuffle (`fingerprint2.py`). Every other object avoids immediate repetition at 0.43–0.83 of expectation. The Staff sits at 0.47 and, uniquely, is depressed at lag 2 as well (0.80). Removing glyph 76 (24.3 % of the Staff's tokens after the divider 999 is excluded, against 0.6 % of the rest of the corpus) returns it to the corpus range at lag 1 (0.85) and lag 2 (1.25). The Staff's statistical isolation on this axis is an artefact of one very frequent glyph, and Fischer's triads leave no trace at the period they predict: lag 3 is 0.94 as encoded and 1.09 with 76 removed.

References

T. S. Barthel. *Grundlagen zur Entzifferung der Osterinselschrift.* Cram, de Gruyter, Hamburg, 1958.

A. Davletshin. Numerals and phonetic complements in the Kohau Rongorongo script of Easter Island. *Journal of the Polynesian Society*, 121(3):243–274, 2012. `https://doi.org/10.15286/jps.121.3.243-274`

A. Davletshin. Allographs, graphic variants and iconic formulae in the Kohau Rongorongo script of Rapa Nui (Easter Island). *Journal of the Polynesian Society*, 126(1):61–92, 2017.

A. Davletshin. The script of Rapa Nui (Easter Island) is logosyllabic, the language is East Polynesian: evidence from cross-readings. *Journal of the Polynesian Society*, 131(2):185–220, 2022. `https://doi.org/10.15286/jps.131.2.185-220`

S. R. Fischer. Preliminary evidence for cosmogonic texts in Rapanui's rongorongo inscriptions. *Journal of the Polynesian Society*, 104(3):303–321, 1995.

J. B. M. Guy. On the lunar calendar of tablet Mamari. *Journal de la Société des Océanistes*, 91:135–149, 1990.

Hamilchin. `indus-cipherable`: digitisation of Mahadevan (1977) and the EBUDS set. `https://github.com/Hamilchin/indus-cipherable`, accessed August 2026.

P. Horley. Allographic variations and statistical analysis of the Rongorongo script. *Rapa Nui Journal*, 19(2):107–116, 2005.

P. Horley. *Rongorongo: Inscribed Objects from Rapa Nui.* Rapanui Press, Viña del Mar, 2021. (Basic-glyph catalogue of ∼130 signs; the Barthel→Horley map used here is the one distributed with `rongopy`, `https://github.com/jgregoriods/rongopy`.)

E. Korovina. A computational evaluation of syllabic hypotheses for Rongorongo: evidence from n-gram analysis. In *Proceedings of the Fourth Workshop on Language Technologies for Historical and Ancient Languages (LT4HALA 2026) @ LREC 2026*, pages 177–183, Palma, 2026. `https://aclanthology.org/2026.lt4hala-1.16/`

I. Mahadevan. *The Indus Script: Texts, Concordance and Tables.* Archaeological Survey of India, New Delhi, 1977.

[3] We have read Sproat's pages directly; the earlier criticisms are taken from his summary and from Pozdniakov and Pozdniakov [2007].

G. A. Miller. Note on the bias of information estimates. In *Information Theory in Psychology*, pages 95–100. Free Press, 1955.

A. Nair. How non-linguistic is the Indus sign system? A synthetic-baseline scorecard. arXiv:2604.17828, April 2026.

K. Pozdniakov and I. Pozdniakov. Rapanui writing and the Rapanui language: preliminary results of a statistical analysis. *Forum for Anthropology and Culture*, 3:3–36, 2007.

R. P. N. Rao, N. Yadav, M. N. Vahia, H. Joglekar, R. Adhikari, and I. Mahadevan. Entropic evidence for linguistic structure in the Indus script. *Science*, 324(5931):1165, 2009.

R. P. N. Rao, N. Yadav, M. N. Vahia, H. Joglekar, R. Adhikari, and I. Mahadevan. Entropy, the Indus script, and language: a reply to R. Sproat. *Computational Linguistics*, 36(4):795–805, 2010.

R. Sproat. Ancient symbols, computational linguistics, and the reviewing practices of the general science journals. *Computational Linguistics*, 36(3):585–594, 2010.

R. Sproat. A statistical comparison of written language and nonlinguistic symbol systems. *Language*, 90(2):457–481, 2014.

R. Sproat. Approximate string matches in the rongorongo corpus. `https://richardsproat.com/ror/`, 2026, accessed August 2026.

B. K. Wells. *The Archaeology and Epigraphy of Indus Writing*. Archaeopress, Oxford, 2015. With an appendix by A. Fuls. Wells's own published sign count is about 676; the Interactive Corpus of Indus Texts (ICIT), the online sign list he maintains separately with Fuls, currently sits at about 702.